SYSTEM DESIGN DOCUMENT

Regulated Evidence Intelligence Platform

Body-Worn Camera Audit | Marston Holdings

Version 5.0 - Architecture-review | AWS + NVIDIA DeepStream

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Version	**5.0 - Architecture-review ready: duplicates removed, all zones in sequence, new flowcharts**
Status	**Draft - For review**
Cloud platform	AWS | NVIDIA DeepStream 7.0+ · Riva Parakeet · NeMo · Triton
Classification	**Proprietary & Confidential**

1. Marston Holdings - Organisation and Enforcement Context

1.1 Who Marston Holdings Are

Marston Holdings is the UK's largest civil enforcement and debt recovery group. Founded in 1999 and headquartered in Staffordshire, the business has grown through acquisition to become the dominant operator in the enforcement sector, processing recoveries on behalf of local authorities, central government, and courts across England and Wales.

1.2 Business Divisions

Division	What they do
NSL Services	Civil Enforcement Officers (CEOs) - parking, bus lanes, moving traffic contraventions. Operates on behalf of 280+ local councils. Largest CEO operation in the UK.
Marston Recovery	Enforcement agents - recovery of council tax, business rates, CCJs, magistrates court fines, High Court writs. Regulated under Taking Control of Goods Regulations 2013.
Videalert	CCTV and ANPR enforcement technology - operates enforcement cameras for bus lanes, moving traffic, and school streets on behalf of local authorities.
Project Centre	Traffic Regulation Order (TRO) design and implementation - creates the legal framework that underpins enforcement.
ParkTrade / Scott & Co	Road user charging and Scottish enforcement - extends the group's reach into tolling and Scottish civil debt recovery.

1.3 Scale of Operations

- ~4,000 field agents deployed daily across England and Wales
- 280+ local authority clients including Transport for London (TfL), DVLA, HMRC, and Ministry of Justice
- Over £850 million recovered annually across all enforcement streams
- Estimated 6,000 –10,000 doorstep visits every working day

1.4 The Three Enforcement Stages

Enforcement visits follow a statutory three-stage process. The SOP, permitted actions, and scoring weights are different at each stage.
(This is why enforcement_stage is a required metadata field in Zone A and a routing key in Zone B.)

Stage	What happens	SOP implications for this platform
Compliance stage	Notice of enforcement sent. 7 clear days before any visit. Debtor can pay or contact agent.	No field visit at this stage. BWC not activated. Platform processes enforcement and sale stage clips only.
Enforcement stage	First visit. Agent can: list goods for potential seizure (controlled goods agreement), accept payment, or abort if vulnerability detected.	Primary platform scope. All SOP steps apply - identity, debt amount, payment options, vulnerability assessment, peaceful entry. High vulnerability detection importance.
Sale stage	Second visit (if first was unsuccessful). Goods previously listed can now be removed and sold.	Goods inventory SOP step becomes mandatory and weighted highest. Vulnerability re-assessment required. Resident may have become more distressed since first visit.

1.5 Debt Collection Types and Enforcement Context

Marston agents collect multiple categories of debt, each with different legal powers, SOP requirements, and vulnerability/escalation risk profiles.

(This directly determines how the SOP configuration engine (Zone B) routes clips to the correct checklist, and why the platform requires multiple use cases rather than a single generic SOP)

Debt type	Entry rights	Enforcement stages	BWC risk profile
Council tax	Peaceful entry only	All three stages	High vulnerability risk - residential, often distressed households
Business rates	Can force entry to commercial premises	All three stages	Commercial context - lower vulnerability, different SOP at levy stage
Parking / PCN	Peaceful entry only	Compliance + enforcement	High volume, lower value. Residents often dispute validity of PCN.
ULEZ / traffic contraventions	Peaceful entry only	Compliance + enforcement	Frequently contested. TfL client - high media scrutiny.
Magistrates court fines	Statutory powers - can force entry with authority	Enforcement + sale	Higher conflict risk. Criminal penalty context - agents hold statutory authority.
County Court Judgments (CCJ)	Peaceful entry - return rights after controlled goods agreement	Compliance + enforcement	Multi-visit complexity. Controlled goods agreement changes subsequent visit SOP.

High Court writs (HCEO)	Can force entry to commercial premises; peaceful entry residential first visit	Enforcement + sale	Highest legal authority. Commercial forced entry is lawful - alert system must not flag as breach.
Child maintenance	Peaceful entry only	All three stages	Highest emotional volatility. Both parties often distressed. Vulnerability assessment critical.
HMRC / DVLA warrants	Specific statutory powers	Variable by warrant type	Statutory authority - highest compliance scrutiny from regulators.

Alert system note	High Court writ enforcement against commercial premises may lawfully involve forced entry. The alert tier system flags forced entry only when it occurs outside the authorised legal context. The use-case router assigns HCEO commercial writs to a separate use case with adjusted alert rules - forced entry is not flagged as a breach for this use case.

1.6 The Body-Worn Camera Programme

Camera hardware - Body Lens

Characteristic	Detail and platform implication
Recording activation	Manual activation by agent. LED indicator visible to resident. Verbal announcement required - this is SOP Step 0, the first compliance check in every use case.
Audio quality	Single integrated microphone. No directional or noise-cancelling hardware. Primary audio challenge for ASR - outdoor visits suffer from wind, traffic, competing voices. Zone A2 audio pre-processing exists specifically to address this.
GPS tagging	Each clip GPS-tagged at start and end. Enables temporal compliance checking (visit time) and geographic routing to the correct regional SOP configuration.
Battery / storage	Full shift coverage (8–12 hours). On-device encrypted storage with tamper-evident seal. Locked until docked.

Upload and storage - Smart Shuttle and Huddle

Smart Shuttle	The mobile upload application on agents' smartphones. When the agent docks their Body Lens, Smart Shuttle detects new clips and uploads over 3G/4G/5G to Huddle with automatic retry, encryption in transit, and metadata tagging (officer_id, device_id, timestamp). Smart Shuttle is the upstream data source for this platform. (Ingestion assumes API or export access to Smart Shuttle or Huddle. This is a pre-PoC critical dependency - if API access is not granted, a manual footage export workflow is the fallback during PoC.)

Huddle	Marston's encrypted video management platform. Stores, indexes, and controls access to all BWC footage. Current retention: 45 days for non-evidential footage. (This platform does not replace Huddle - it analyses footage and produces intelligence alongside it. The API integration is subject to commercial agreement with Marston's technology team.)

The gap this platform fills

Under current practice, footage is reviewed reactively - fewer than 5% of clips ever watched. An agent could deliver hundreds of visits with missed SOP steps, escalating behaviour, or vulnerability failures and remain invisible until a formal complaint arrives. By then the 45-day retention window may have passed. The ECB now commissions random BWV sampling of accredited firms - Marston cannot know what the ECB will find before they find it. This platform converts the BWC programme from a reactive evidence archive into a proactive intelligence system.

2. Executive Summary

This document defines the System Design for a Regulated Evidence Intelligence Platform (REIP) - an AI pipeline that ingests daily BWC footage, automatically analyses every enforcement visit against three operational questions, and surfaces actionable, explainable intelligence to supervisors and compliance teams.

IS	Regulated evidence intelligence processing - every clip is a legal artefact tied to a debt enforcement action governed by Taking Control of Goods Regulations 2013, under ECB oversight.

IS NOT	Video analytics, surveillance, or a dashcam review tool. The platform treats footage as legal evidence. AI flags; humans decide. All consequential decisions are taken by people.

2.1 The Three Operational Questions

Q	Question	Outputs
Q1	**Was the visit procedurally compliant?**	Identity confirmed · recording announced · legal authority explained · debt amount stated · payment options offered · vulnerability assessed · peaceful entry respected · goods inventory conducted · visit within permitted hours
Q2	**Did the interaction present escalation or safety risk signals?**	Raised voices (origin) · threat/discriminatory language · distress indicators · provocation patterns · proximity violations · lunging behaviour · objects thrown · exit obstruction · hostile 3rd party · camera obstruction · de-escalation outcome
Q3	**What was the visit outcome?**	Payment collected · payment plan agreed · entry refused · goods levied · aborted (vulnerability) · no contact · revisit required · enforcement escalated
→	**Derived: compliance review priority**	Composite indicator computed from Q1+Q2. Internal schema field: compliance_review_priority (low/medium/high/critical). Dashboard display label: Litigation Risk Indicator. Not a standalone question - output of the scoring engine.

3. Detection Modality - Audio/Text vs Video

Every signal across Q1, Q2, and Q3 is detected by audio/text analysis, video/image analysis, or both. This table defines the modality and the AI engine responsible. ★ Phase 2 signals require labelled training data not yet available.

Q	Signal	Audio / Text	Video / Image	Engine
Q1	Identity confirmed	Primary	Supporting	Riva ASR + NeMo NLP + DeepStream
Q1	Recording announced	Primary	- (LED check only)	Riva ASR + NeMo NLP
Q1	Legal authority explained	Primary	-	NeMo NLP phrase detection
Q1	Debt amount stated (cross-ref case data)	Primary	-	Riva ASR + NeMo entity extraction
Q1	Payment options offered	Primary	-	NeMo NLP keyword match
Q1	Vulnerability assessed	Primary	Supporting	NeMo NLP + DeepStream (age/disability cues)
Q1	Peaceful entry respected	Supporting	Primary	DeepStream (posture, proximity, door scene)
Q1	Goods inventory conducted (levy stage)	Primary	Supporting	NeMo NLP + DeepStream (document visible)
Q1	Visit within permitted hours (06:00–21:00)	Metadata only	Metadata only	Clip timestamp - metadata check only
Q2	Raised voices - with origin attribution	Primary	-	Riva ASR dB tracking + NeMo causal chain
Q2	Threat language (direct, implied, conditional)	Primary	-	NeMo NLP threat classifier
Q2	Discriminatory language	Primary	-	NeMo NLP hate-speech classifier
Q2	Resident distress indicators	Primary	Supporting	Riva ASR (voice tremor) + DeepStream
Q2	Provocation patterns (agent-triggered)	Primary	-	NeMo NLP causal sequence analysis
Q2	Proximity violations ★ Phase 2	-	Primary	DeepStream - requires labelled training data
Q2	Lunging / objects thrown ★ Phase 2	-	Primary	DeepStream action classifier - Phase 2
Q2	Camera obstruction / deactivation	-	Primary	DeepStream - black frame / angle change
Q2	De-escalation attempt and outcome	Primary	Supporting	NeMo NLP + DeepStream posture
Q3	Visit outcome classification	Primary	Supporting	NeMo NLP + DeepStream scene end

4. System Architecture

4.1 Architecture Overview

The platform is organised into seven functional zones. Each zone has a single responsibility, clear inputs, and clear outputs. The architecture is AWS-first, NVIDIA GPU-accelerated, and designed so that SOP checklists, scoring weights, and use-case configurations are stored as data - not code - enabling updates without deployment.

Figure 1 - Layered system architecture

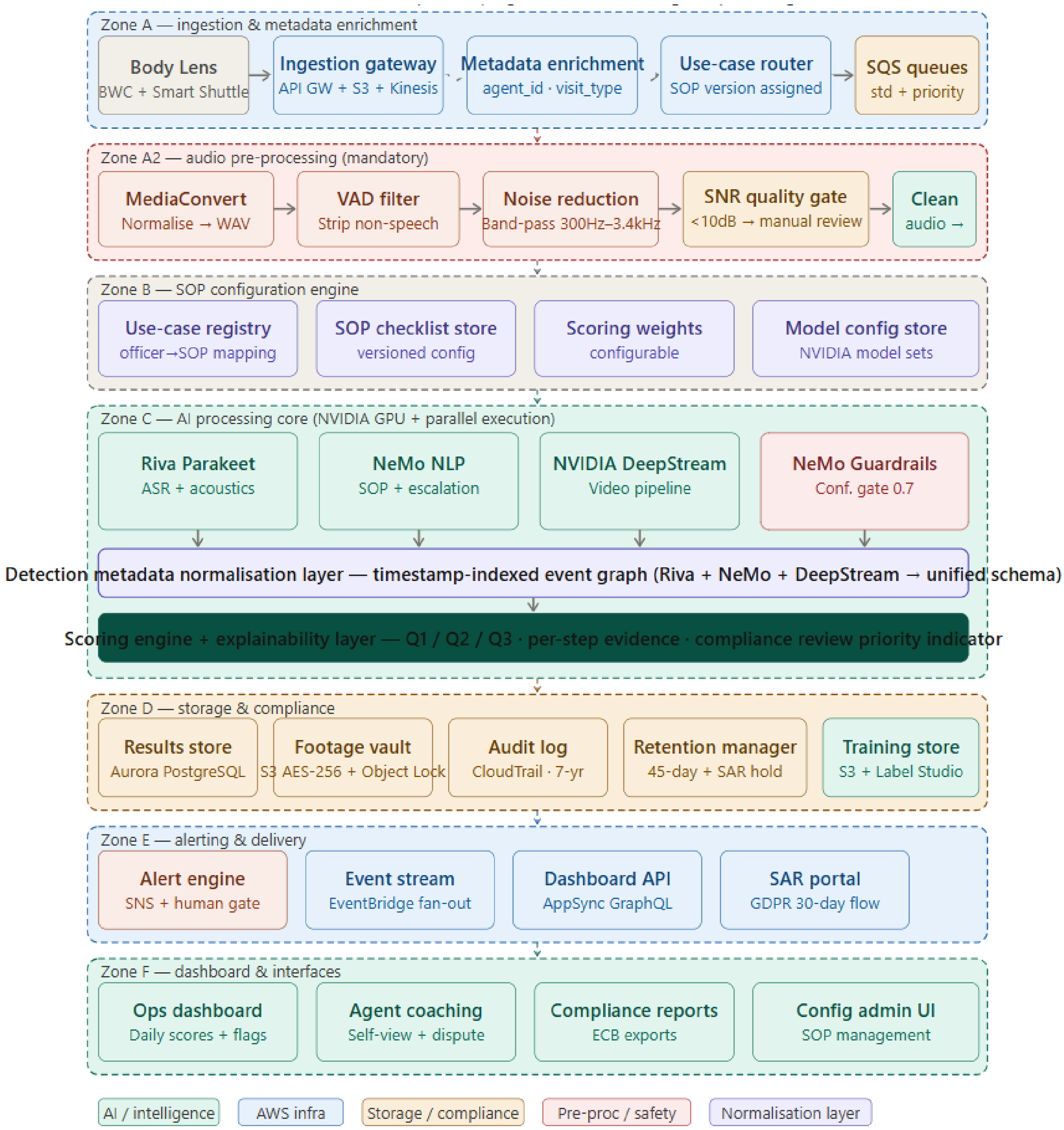

Figure 1: All seven processing zones from field ingestion (Zone A) to dashboard output (Zone F). Zone A2 (audio pre-processing) and the detection metadata normalisation layer inside Zone C are new in v5.

4.2 Zone A - Ingestion & Metadata Enrichment

Zone A is the single authenticated entry point for all footage. Every clip is tagged with case metadata on arrival, validated, and routed to the correct SOP pipeline before any AI analysis begins.

Component	AWS service	Responsibility
Body Lens / Smart Shuttle	Marston proprietary	BWC device + 3G/4G/5G upload with auto-retry. Tags clips with case_id and officer_id. API access subject to Huddle integration agreement.
Ingestion gateway	API Gateway + S3 + Kinesis	Authenticated entry. Batch clips land in S3; priority clips enter Kinesis Video Streams. Returns clip_id and s3_uri.
Metadata enrichment	Lambda + DynamoDB	Joins clip metadata with case data: officer_type, visit_type, client_council_id, region, enforcement_stage.
Use-case router	Lambda + SQS	Assigns use_case_id + sop_version from Zone B config. Standard or priority SQS queue. Unmatched clips → manual_review queue with supervisor alert.

4.3 Zone A2 - Audio Pre-Processing

Literature review confirmed outdoor enforcement audio can degrade ASR word error rate from ~8% to over 90% without pre-processing. This zone is mandatory for every clip.

Step	Technology	Function
1 - Normalise	AWS Elemental MediaConvert	Convert to WAV 16kHz mono. Extract audio track separately for faster processing. ** WAV (Waveform Audio File Format) file is a high-quality, uncompressed audio format widely used for professional audio production and archiving.
2 - VAD	WebRTC VAD	Voice Activity Detection: removes silence, door slams, traffic bursts, barking. Produces speech-only segments.
3 - Noise reduction	Band-pass filter 300Hz–3.4kHz	Isolates human speech frequency range. Attenuates wind, rain, traffic below 300Hz and above 3.4kHz.
4 - SNR gate	Signal-to-noise measurement	SNR ≥ 10dB → proceed to Riva ASR. SNR < 10dB → flag audio_quality: poor, route to manual review. No unreliable transcript produced.

4.4 Zone B - SOP Configuration Engine

Zone B defines what a compliant visit looks like for every enforcement type. All configuration is stored in DynamoDB and editable via the admin UI - updating an SOP requires no code deployment.

- Use-case registry - maps officer_type + visit_type + enforcement_stage + client_id to a specific SOP checklist and model set
- SOP checklist store - versioned list of required steps, detection keywords, weights, and order rules
- Scoring weights - configurable per use case; weights must sum to 100; required steps auto-fail if absent
- Model config store - specifies which NVIDIA model set and confidence thresholds apply per use case
- Annotation tooling - Label Studio / AWS Ground Truth for building training datasets (Phase 2)

SOP routing matrix - example

The routing table shows how metadata from Zone A resolves to a specific use case and SOP version:

officer_type	visit_type	enforcement_stage	client_id	→ use_case_id + sop_version
bailiff	residential_debt	enforcement	tfl_001	bailiff_enforcement_v2 · SOP_2025-02
bailiff	residential_debt	sale	tfl_001	bailiff_sale_v2 · SOP_2025-02
ceo	pcn_enforcement	enforcement	council_042	ceo_pcn_v3 · SOP_2025-01
hceo	commercial_writ	enforcement	private_001	hceo_commercial_v1 · SOP_2025-03 (forced entry permitted)
bailiff	child_maintenance	enforcement	cms_001	bailiff_cms_v1 · SOP_2025-04 (enhanced vuln. weighting)

4.5 Zone C - AI Processing Core

Zone C runs four AI engines in stream-based parallel execution, then fuses their outputs through a normalisation layer and scoring engine. All inference runs on NVIDIA GPU instances (EC2 G4dn/P4d/P5) via Triton Inference Server.

4.5.1 NVIDIA Riva Parakeet ASR - speech and acoustics

Riva Parakeet handles both the transcript and the raw acoustic feature extraction. It produces speaker-attributed utterances with acoustic signal data. NeMo NLP then consumes this output for language classification - NeMo is a language model, not an audio processor.

Attribute	Detail
Model	Riva Parakeet-CTC - Fast Conformer encoder, 2.4× faster than standard Conformer. UK-EN primary. Welsh visits routed to AWS Transcribe Welsh model.
Transcript output	[{ speaker: 'agent'|'resident'|'third_party', text, start_ms, end_ms, confidence }]
Acoustic output	Per utterance: { db_level, pitch_mean_hz, pitch_variance, speech_rate_wpm, pause_before_ms }. Computed on speaker-separated audio - not contaminated by opposing speaker.
Edge cases	Overlapping speech: diarisation confidence flag. Low confidence utterances not used in SOP detection. Speaker count > 2: third_party_detected flag.

4.5.2 How tone, pitch, and volume are analysed

The most important signals in an enforcement visit are often not what was said - it is how it was said. An agent who says 'I understand your concern' with a dismissive, clipped delivery is behaving very differently from one who means it. The platform extracts three acoustic dimensions per utterance:

Dimension	What we measure	What it tells us
Volume / amplitude	Decibel (dB) level per utterance per speaker, tracked over time	Volume spike on agent channel before resident = agent-initiated escalation. Gradual volume decrease after a peak = de-escalation in progress. Sustained high = unresolved conflict.
Pitch / frequency	Fundamental frequency (F0) contour - rise and fall of vocal pitch in Hz	High sustained pitch = stress or distress. Flat monotone in agent = disengagement. Trembling or breaking pitch = resident fear or mental health distress.
Speech rate	Words per minute, pause duration, interruption frequency	Very slow deliberate speech = coercive pacing. Interruption pattern = dominance signal / procedural justice risk. Long pauses after agent demand = resident distressed.

4.5.3 NVIDIA NeMo NLP - language understanding

NeMo operates on the Riva transcript plus acoustic features. It runs two streams simultaneously: SOP compliance detection and escalation signal detection.

Signal category	Detection method	Output
SOP step keywords	NeMo checks transcript against SOP checklist from Zone B. Searches for keyword patterns per speaker channel in expected timing window.	Per step: found, confidence, timestamp_ms, quote

Threat language	NLP classifier: direct ('I will...'), conditional ('unless...then...'), implied ('you won't like...'). Both channels.	threat_detected, threat_type, speaker, confidence, timestamp_ms, quote
Discriminatory language	Hate speech classifier: race, religion, gender, disability, age. UK-specific terms. Very high sensitivity - any detection flags immediate tier-1 alert regardless of confidence score.	discrimination_flag, type, speaker, severity, timestamp_ms. Auto: ecb_reportable=true
Provocation patterns	Sequence: agent utterance (tone: dismissive/aggressive) followed within 15 seconds by resident volume spike. Causal chain confirms link.	provocation_by_agent, trigger_utterance, resident_response, confidence
Escalation origin	Causal chain analysis: agent statement → resident spike = agent-triggered. Confidence > 0.85 required to attribute; otherwise output origin: ambiguous.	escalation_origin: agent | resident | mutual | ambiguous
De-escalation	Calming language after escalation event. Timing matters - within 60 seconds scores higher.	deescalation_attempted, timing_seconds, outcome: successful/failed/partial

4.5.4 NVIDIA DeepStream - video processing backbone

DeepStream processes clips as a continuous stream rather than static 1fps frame samples, enabling detection of events that unfold across time - camera being gradually obscured, posture shifting, person entering scene.

Figure 2 - NVIDIA DeepStream inference lifecycle

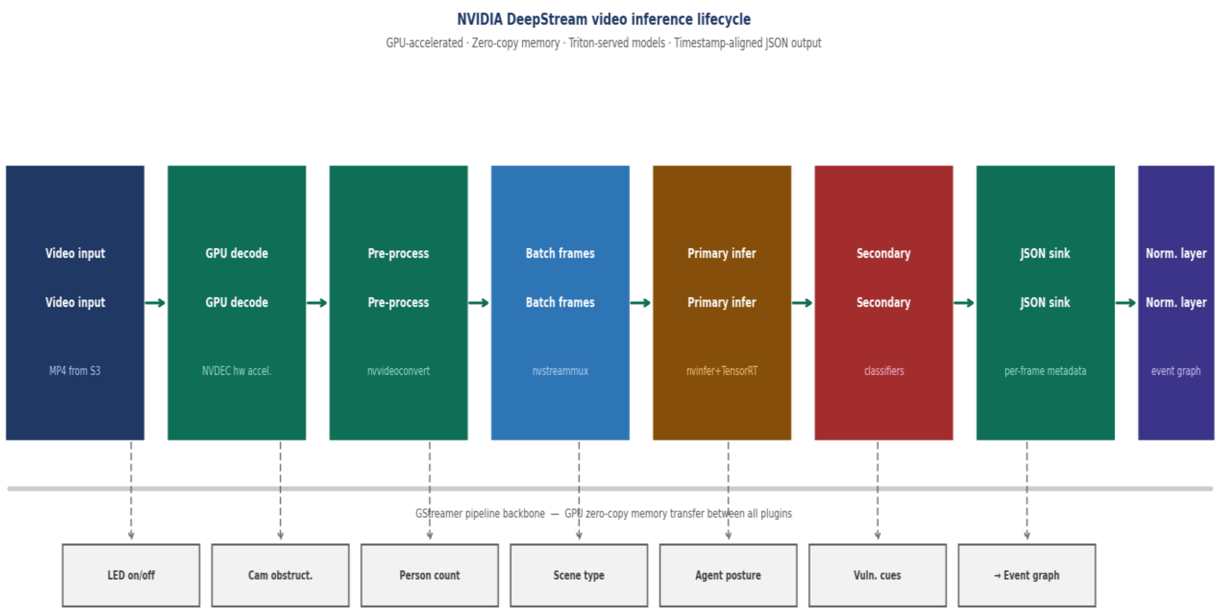

Figure 2: GStreamer pipeline - GPU decode (NVDEC) → pre-process → batch frames (nvstreammux) → primary inference (nvinfer+TensorRT) → secondary classifiers → timestamp-aligned JSON metadata → detection normalisation layer.

Detection (PoC scope)	Method	Platform use
BWC LED status	Continuous LED on/off detection - not just clip start	LED off while case open: camera_obstruction flag, tier-1 escalation
Camera obstruction	Solid black frame detection + sudden angle change + hand-over-lens texture analysis	Immediate tier-1 alert. ecb_reportable=true. Highest-risk single signal.
Person count	Person detection model: counts individuals per second, tracks entry/exit	third_party_detected if count > 2. Child present = automatic vulnerability flag.
Agent posture	Pose estimation: neutral vs confrontational vs blocking doorway	posture_flag if confrontational > 10 seconds. Feeds peaceful_entry_respected SOP check.
Vulnerability visual cues	Secondary classifiers: elderly_indicators, disability_aids, child_present	vulnerability_visual_cues[]. Cross-modal check with NeMo required - vision alone does not confirm.
Scene classification	Scene type at visit start: residential door, communal, outdoor, commercial	Validates use-case routing. Mismatch (e.g. commercial scene on residential SOP) triggers use_case_mismatch flag.

Phase 2 physical signal detections (proximity violations, lunging, objects thrown) require labelled training data and 5fps+ frame rate - excluded from PoC scope.

4.5.5 Detection metadata normalisation layer

This layer sits between the three AI engines and the scoring engine. Its purpose is to convert heterogeneous outputs - frame-level detections from DeepStream, utterance-level detections from Riva and NeMo - into a single unified, timestamp-indexed event graph that the scoring engine can consume consistently.

Why this layer exists	Without normalisation, the scoring engine would need to understand DeepStream JSON format, Riva transcript format, and NeMo classification format simultaneously. The normalisation layer abstracts this - the scoring engine always receives the same schema regardless of which model produced which signal.

Function	Detail
Schema alignment	Converts DeepStream frame-level JSON, Riva utterance objects, and NeMo classification outputs into a common event schema: { event_type, timestamp_ms, end_ms, speaker, confidence, source_model, payload }
Timestamp indexing	All events sorted into a single chronological timeline. Enables causal chain analysis - which event preceded which. Critical for escalation origin attribution.
Confidence propagation	Tracks confidence per source: audio_confidence (Riva), vision_confidence (DeepStream), nlp_confidence (NeMo). Feeds into overall_confidence calculation in the scoring engine.
Cross-modal alignment	Matches vision events to corresponding audio events by timestamp window. Required for cross-modal contradiction gate in NeMo Guardrails.

Overall confidence formula:

```
overall_confidence = min(audio_conf * 0.5 + vision_conf * 0.3 + nlp_conf * 0.2)
```

The minimum function ensures that a single low-confidence modality pulls the overall score down - preventing a high-confidence text result from masking a poor-quality video analysis.

4.5.6 NeMo Guardrails - output safety gate

NeMo Guardrails prevents unreliable AI outputs from reaching supervisors as facts. Literature review identified false positives as the primary reason AI BWC programmes lose trust and get cancelled.

- Confidence threshold gate - any detection below 0.7 confidence is surfaced as uncertain, not a definitive flag
- Hallucination check - ASR utterances with no acoustic evidence are removed before scoring
- Cross-modal contradiction gate - vision and NLP must agree before a vulnerability flag is raised
- Escalation origin gate - attribution only when causal chain confidence > 0.85; otherwise ambiguous

Why 0.85 threshold matters	Incorrectly attributing escalation origin to an agent who did not start it could result in unfair disciplinary action. The system never attributes when confidence is low. Ambiguous cases go to supervisor review with the full acoustic evidence visible.

4.5.7 Scoring engine and explainability layer

Combines all engine outputs into a single structured result. Every score includes per-step evidence so agents and supervisors understand exactly why a score was given.

```
{ clip_id, agent_id, visit_id, use_case_id, sop_version,
 q1_score: 0-100,
 q1_breakdown: [{ step_id, found, confidence, timestamp_ms, quote }],
 q2_escalation_level: none|low|medium|high,
 q2_evidence: [{ type, speaker, timestamp_ms, clip_ref, confidence }],
 q3_outcome: enum, q3_confidence: float,
 compliance_review_priority: low|medium|high|critical,
 ecb_reportable: bool, audio_quality: ok|poor, overall_confidence: float }
```

Scoring rules

Condition	Scoring rule
Required SOP step not detected	Automatic deduction regardless of weights. Score cannot exceed 60 if any required step is missing. Two+ required steps missed → score cannot exceed 50.
Optional SOP step not detected	Weighted deduction applied. Step weight subtracted from base score of 100.
Camera obstruction confirmed	Immediate q2_escalation_level = high. ecb_reportable = true. compliance_review_priority = critical. Supervisor tier-1 alert triggered regardless of other scores.
Discriminatory language (any speaker)	compliance_review_priority = critical. ecb_reportable = true. Immediate tier-1 alert. No confidence threshold applied - any detection triggers.
Vulnerability detected + no SOP vulnerability assessment	Automatic tier-2 alert. Compliance_review_priority elevated to high regardless of Q1 score.
Visit outside permitted hours	visit_time_compliant = false. Automatic compliance flag. Regulatory breach regardless of visit quality.

4.6 Zone D - Storage and Compliance

Zone D is the legal and operational spine of the platform. Every piece of footage, every AI result, every access event, every deletion is stored, tracked, and auditable here. Zone D serves four purposes: evidence preservation, intelligence persistence, regulatory compliance, and model improvement.

Results store - Aurora PostgreSQL

Primary operational database. Stores every visit result record, transcript as a first-class legal document, per-step score breakdown, and agent rolling summary views. Designed for the query patterns the dashboard requires: by agent, by date range, by flag type, by score band, by outcome.

- Schema: clip_id (PK), agent_id, visit_date, use_case_id, sop_version, q1_score, q2_escalation_level, q3_outcome, compliance_review_priority, ecb_reportable, audio_quality, overall_confidence, processing_timestamp
- Transcript table: separate table per clip. Each row is one utterance: speaker, text, start_ms, end_ms, tone, acoustic features. Stored as a primary legal evidence artefact - not merely a score input
- Score breakdown table: one row per SOP step per clip. step_id, found, confidence, timestamp_ms, quote
- Agent summary: materialised view updated nightly. Per-agent rolling averages for 30/60/90-day windows

Footage vault - Amazon S3 with Object Lock

- AES-256 server-side encryption. Each clip encrypted with a unique KMS-managed key
- S3 Object Lock (COMPLIANCE mode) - prevents deletion of any clip with active SAR or evidential hold. Not even root account can delete before lock expires
- Folder structure: s3://bwc-vault/{council_id}/{agent_id}/{year}/{month}/{clip_id}.mp4
- Lifecycle rules: non-evidential clips moved to S3 Glacier at day 30, deleted at day 45. Evidential and SAR-held clips: Object Lock overrides lifecycle
- All footage access via time-limited pre-signed URLs (15-minute expiry) generated by Lambda after RBAC check - no direct bucket access for any user role

Immutable audit log - AWS CloudTrail

- Events logged: clip upload, AI processing, result written, any user access, coaching note, score override, score disputed, deletion, SAR received, SAR hold placed/released
- Log integrity: CloudTrail log files are hash-chained. Any modification invalidates the chain. Replicated to a separate AWS account preventing single-account compromise
- Retention: 7 years minimum - exceeds Limitation Act 1980 standard period for civil claims

Retention manager and SAR hold engine

Condition	Action
Age < 45 days AND no hold	No action. Next check tomorrow.
Age >= 45 days AND no hold	Delete footage from S3, transcript from Aurora, score record flagged footage_deleted. Deletion logged to CloudTrail.
Any age + SAR hold	No deletion. S3 Object Lock enforces at storage layer. DPO must release hold through four-eyes authorisation.
Any age + evidential hold	No deletion. Requires two-person authorisation by Compliance or Legal team to release.

4.7 Zone E - Alerting and Delivery

Zone E delivers intelligence to the right person at the right time. Core principle: AI flags; humans decide. No alert causes automated action against an agent. All consequential decisions are taken by people after reviewing evidence.

Three-tier alert system

Tier	Timing	Trigger and routing
Tier 1 - Immediate	Within 5 minutes	Camera obstruction · discriminatory language · forced entry (non-HCEO) · ECB-reportable combination. Push notification to supervisor + compliance manager + DPO. Clip locked. Supervisor must acknowledge within 2 hours.
Tier 2 - Same day	By 14:00	Q2 escalation HIGH · vulnerability flag + SOP failure · compliance_review_priority = critical · threat language (agent, high confidence). Dashboard alert + email. Review expected within 24 hours.
Tier 3 - Morning digest	By 07:00	All other flagged clips. Q1 below threshold. Q2 medium. Any SOP step missed. Q3 anomaly. Processed as part of daily batch review.
Pattern alert - weekly	Monday morning	Q1 trend declining > 10pts over 4 weeks · escalation rate above regional average · specific SOP step consistently missed. Routing: regional manager + HR. Coaching conversations, not disciplinary triggers.

EventBridge fan-out (concurrent, not sequential)

When Zone C completes scoring, it publishes a clip_processed event to EventBridge. This triggers four concurrent listeners - Zone D writes results, Zone E evaluates alerts, Zone F updates the dashboard, and the SAR retention logic checks holds. Adding a new consumer (e.g. case management integration) requires no change to Zone C.

4.8 Zone F - Dashboard and Interfaces

Zone F is how the platform's intelligence reaches people. Three design principles: no numbers without evidence, supervisors are coaches not judges, agents must be able to see and challenge.

Ops dashboard - supervisor view

Panel	Content and function
KPI summary strip	Five headline numbers: clips processed, average Q1 score, escalation flag count (by tier), vulnerability flags, clips requiring review. Colour-coded vs previous week.
Priority review queue	Ranked clips requiring review today, highest urgency first. Agent, visit time, Q1 score, Q2 flag type, one-line reason. Click to open clip analysis view.
Visit outcome distribution	Bar chart: Q3 outcome breakdown - payment collected, plan agreed, entry refused, aborted, revisit. Tracks against target distribution.
SOP step pass rates	Per-step compliance rate across all clips today. Surfaces systematic training gaps before they become complaints.
Agent score summary	One row per agent: visits, average Q1, flag count, primary outcome, 30-day trend arrow. Click to open coaching view.

Clip analysis view - single visit

Panel	Content and function
Q1/Q2/Q3/Risk header	Four-panel header: headline result per question plus Litigation Risk Indicator. Colour-coded. Each panel shows score and one-line reason.
SOP compliance checklist	Every SOP step with status: pass (green tick), fail (red cross), uncertain (amber warning). Detected transcript quote, timestamp, confidence score. Click to jump to that moment in the transcript.
Visit timeline	Horizontal timeline. Colour-coded event markers: green = SOP detected, red = escalation, amber = warning, blue = de-escalation. Hover for detail. Click to highlight transcript line.
Escalation evidence panel	All Q2 signals: type, speaker, timestamp, confidence, supporting quote. For escalation origin: shows causal chain - what was said before the spike and what happened after.
Transcript panel	Full timestamped, speaker-attributed transcript. Colour-coded highlights: teal = SOP phrase, red = threat/escalation, amber = vulnerability, orange = distress.
Supervisor actions	Coaching note, score override (mandatory reason, logged to audit trail), flag for review, mark as cleared.

Agent coaching view and dispute process

- Agents can log in and see their own scores, transcripts, and per-step breakdown. Cannot access other agents' data.
- 30/60/90-day trend: score history visualised. Highlights consistently weak SOP steps - basis for targeted coaching.
- Positive identification: strong performances surfaced as clearly as flags. Clips where agent handled difficulty well flagged as positive coaching examples.
- Dispute submission: agent submits dispute with reason. Routed to supervisor. Decision logged. Upheld disputes become Phase 3 training labels.

Compliance reports and config admin

- ECB compliance export: formatted for ECB Agent Standards submission. Per-standard compliance rate across reporting period.
- Regional breakdown: compliance rates by council, region, enforcement type. Identifies geographic training gaps.
- Config admin UI: SOP checklist editor, scoring weight configuration, change preview (runs updated config against historical clips before activating).

5. Data Flow

5.1 Parallel vs Sequential Execution Map

The pipeline is not fully sequential. Understanding which stages run in parallel and where they join is essential for estimating throughput and debugging latency.

Phase	Stages and execution model
Sequential	Upload → Metadata enrichment → Audio pre-processing. Each stage must complete before the next begins. Audio pre-processing produces two outputs: clean WAV for Riva, and the normalised MP4 for DeepStream.
Parallel split	Riva ASR, DeepStream video processing, and NeMo NLP run concurrently on the same clip. Riva processes audio → transcript + acoustic features. DeepStream processes video → frame metadata. NeMo is queued and begins as soon as Riva transcript is available.
Join point	Detection metadata normalisation layer waits for all three engines to complete, then merges outputs into the unified event graph. This is the parallel join point.
Sequential	NeMo Guardrails → Scoring engine. These run in sequence on the normalised event graph.
Parallel fan-out	After scoring, EventBridge triggers concurrent writes to: Aurora results store, alert evaluation, dashboard update via AppSync subscription, and SAR retention logic check. These are independent and do not wait for each other.

Figure 3 - Inference execution flow

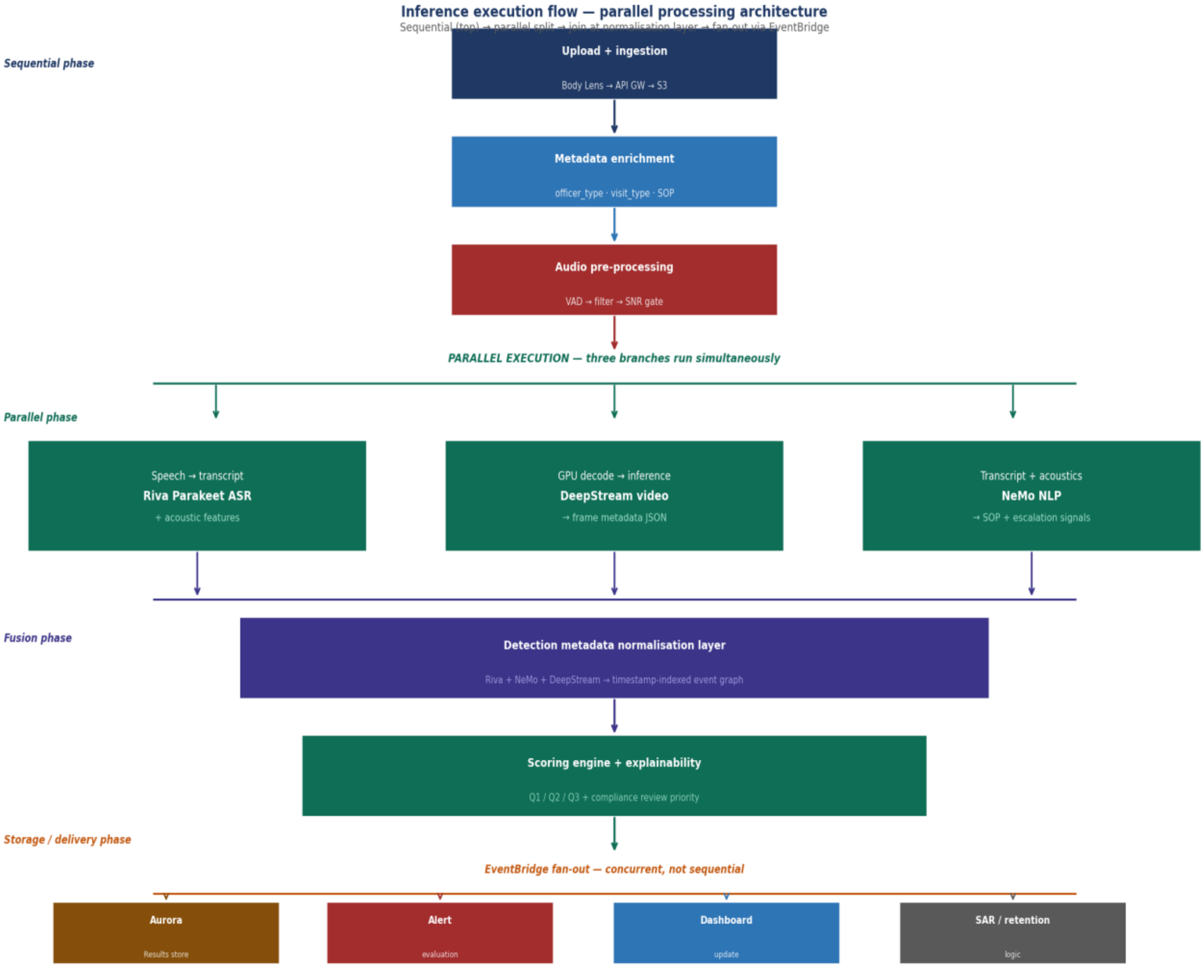

Figure 3: Sequential top section → parallel three-branch inference (Riva / DeepStream / NeMo) → join at detection metadata normalisation layer → scoring → concurrent EventBridge fan-out to storage, alerts, and dashboard.

5.2 Per-Visit Processing Pipeline

Figure 4 - Processing pipeline (upload to result)

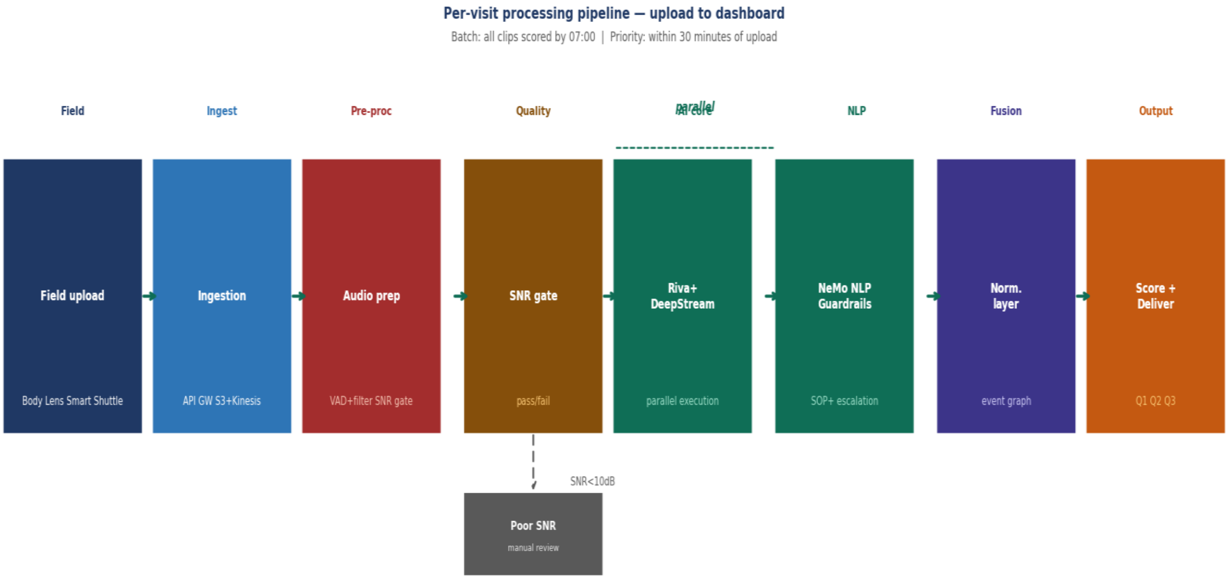

Figure 4: Eight-stage pipeline from Body Lens upload through audio pre-processing, parallel AI analysis, Guardrails gate, Scoring, to Dashboard delivery. Poor-quality audio clips (SNR < 10dB) branch to manual review queue.

5.3 Step-by-Step: Standard Enforcement Visit

#	Zone	What happens
1	A - Ingestion	Agent ends visit. Smart Shuttle uploads clip to API Gateway over 4G. S3 raw bucket receives file. Lambda triggered.
2	A - Enrichment	Metadata Lambda resolves: officer_type=bailiff, visit_type=enforcement_stage, client=London_Borough_X.
3	A - Router	Use-case router assigns use_case_id=bailiff_enforcement_v2, sop_version=2025-02. Job placed in SQS standard queue.
4	A2 - Audio	MediaConvert normalises to WAV. VAD strips silence. Band-pass filter applied. SNR measured at 14dB - passes. Proceeds to Riva.
5	C - Parallel: Riva	8-minute clip transcribed. Agent and resident voices separated. 47 utterances output with timestamps, acoustic features, and confidence scores.
6	C - Parallel: DeepStream	Video processed concurrently: BWC LED on confirmed, 2 persons, no child, no obstruction, agent posture neutral. JSON metadata per frame.
7	C - NeMo NLP	SOP check: steps 0–4 detected, step 5 (payment plan offered) not found. Raised voices at 05:14 - resident-initiated (confidence 0.88).
8	C - Normalisation	Riva + DeepStream + NeMo outputs merged into timestamp-indexed event graph. overall_confidence = 0.82.
9	C - Guardrails	All detections pass 0.7 threshold. Escalation origin confidence 0.88 - attributed to resident. No hallucination flags.
10	C - Scoring	Q1 score: 82 (step 5 missed, −15pts). Q2 escalation: medium. Q3: entry_refused. compliance_review_priority: low.
11	D - Storage (concurrent)	Result written to Aurora. Transcript stored as legal document. Audit log entry created. 45-day retention clock started.

12	E / F - Deliver	Score 82, medium escalation - Tier 3 only (morning digest). Available in supervisor dashboard by 07:00.

Figure 5 - Compliance signal extraction flow

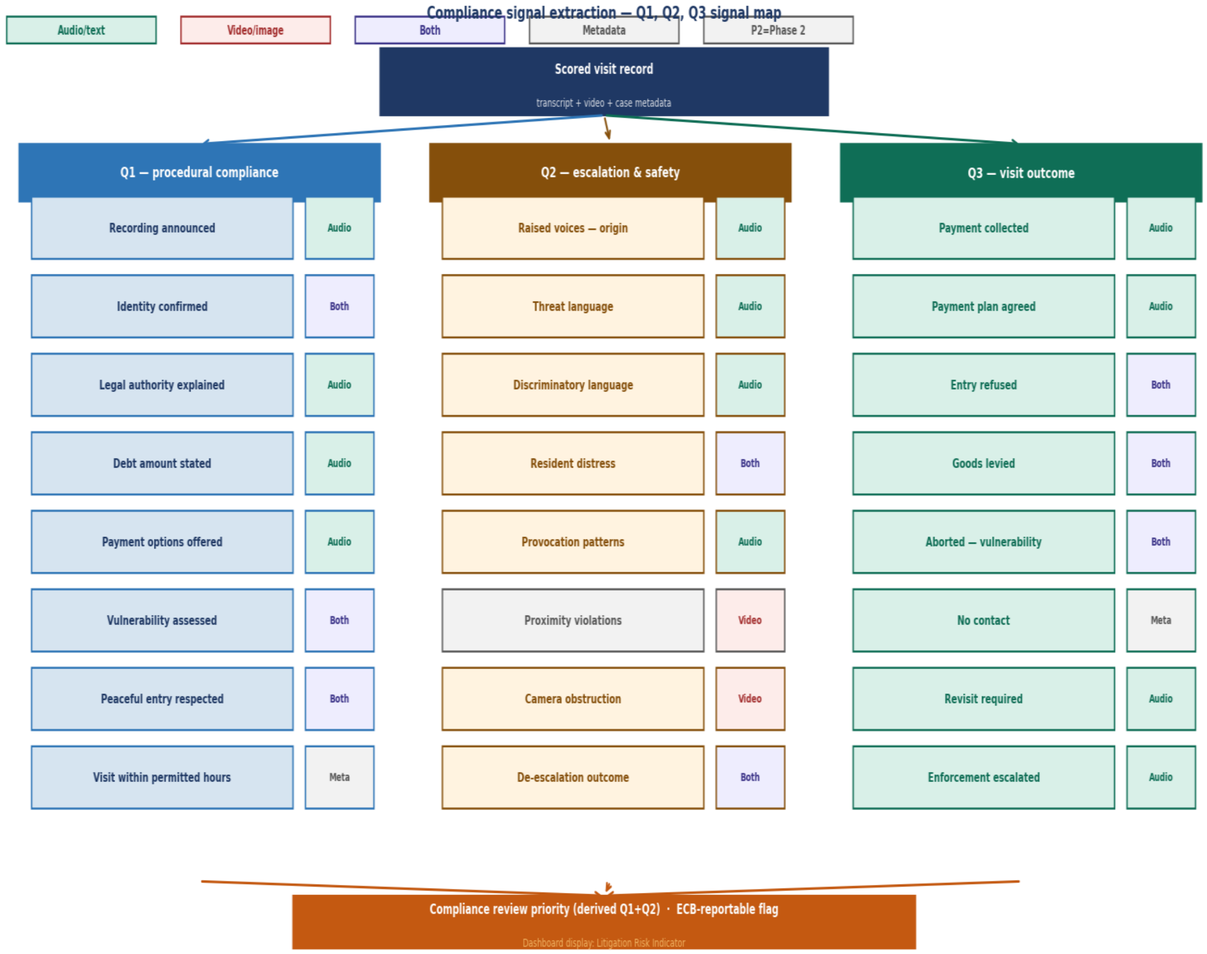

Figure 5: All Q1, Q2, and Q3 signals mapped to their detection source and modality. Phase 2 signals require labelled training data. compliance_review_priority derived from Q1+Q2 by the scoring engine.

6. Technology Stack

6.1 NVIDIA Stack

Component	Role
NVIDIA Riva Parakeet-CTC	Primary ASR. Fast Conformer architecture, 2.4× speed advantage, UK-EN model, speaker diarisation, acoustic feature extraction, confidence scoring.
NVIDIA DeepStream 7.0+	Video processing backbone. GStreamer-based, 40+ GPU-accelerated plugins, zero-copy memory. Runs on AWS EC2 G4dn/P4d/P5 via NGC Docker container.
NVIDIA NeMo Framework	NLP model development and fine-tuning. Phase 1: zero-shot prompting. Phase 2: domain fine-tuned on annotated enforcement audio.
NVIDIA Triton Inference Server	Serves DeepStream vision models, Riva ASR, and NeMo NLP concurrently on GPU. Handles batching, model versioning, and concurrent request management.
NVIDIA NeMo Guardrails	AI output safety layer. Confidence thresholding, hallucination detection, cross-modal contradiction gate.
TensorRT	Model optimisation for DeepStream and Riva - reduces latency and GPU memory footprint.

6.2 AWS Stack

AWS Service	Zone	Usage
API Gateway + S3 + Kinesis	A	Ingestion gateway. Batch to S3; real-time to Kinesis Video Streams.
Lambda	A, C, D, E	Metadata enrichment, routing, scoring engine, retention manager, SAR holds.
SQS	A	Standard and priority processing queues.
EC2 G4dn / P4d / P5 (GPU)	C	NVIDIA T4/A100/H100 GPU. DeepStream Docker from NGC. Auto-start on S3 trigger via CloudWatch event.
Elemental MediaConvert	A2	Video/audio normalisation to WAV 16kHz.
Aurora PostgreSQL	D	Results, transcripts, score breakdowns, agent summaries.
DynamoDB	A, B, D	SOP config, routing tables, hold records, metadata.
S3 + Object Lock	D	Footage vault - AES-256 encrypted, Object Lock prevents deletion during holds.
CloudTrail + CloudWatch	D	Immutable 7-year audit log. CloudWatch triggers GPU instances on new S3 clips.
EventBridge + SNS	D, E	Retention scheduling, concurrent fan-out, alert delivery.
AppSync (GraphQL)	E, F	Real-time dashboard API with live subscriptions. Role-scoped at API layer.
Cognito	F	Authentication and RBAC across all interfaces.

KMS	All	Encryption key management.
Transcribe (Welsh)	C	Welsh-language ASR for Wales-region visits.

7. Security and Compliance

7.1 Regulatory Framework

Regulation	Platform response
UK GDPR Article 35 - DPIA	Data Protection Impact Assessment is mandatory before deployment - systematic monitoring of workers processing sensitive enforcement data meets the high-risk threshold. DPIA must assess proportionality under HRA Article 8 and include agent consultation.
UK GDPR / DPA 2018	DPIA completed before deployment. Lawful basis established. Purpose limitation enforced - footage used only for stated enforcement compliance purpose. Data minimisation applied.
ICO Surveillance Camera Code	12 guiding principles observed. Transparency: agents briefed; LED active. Proportionality: AI flags, humans decide. Full accountability via immutable audit trail.
Human Rights Act Art. 8	Privacy by design. Recording limited to enforcement interactions. No continuous or covert recording. System purpose is compliance verification, not surveillance - proportionality must be documented in DPIA.
ECB Standards (Oct 2024)	SOP checklists aligned with ECB Agent Standards mandatory from Jan 2025. ECB-reportable incident detection built into scoring rules. ECB compliance export in platform.
Taking Control of Goods Regs 2013	SOP steps aligned with statutory requirements per enforcement stage. Visit time check (06:00–21:00). Peaceful entry verification. Vulnerability protections.
CPIA 1996	Transcripts and footage stored with chain of custody suitable for court production.
Welsh Language Act 1993	Welsh ASR routing for Wales-region visits via AWS Transcribe Welsh model.

Note: Zone D contains the full data retention decision tree, SAR hold engine, footage vault access controls, and audit log specification - see Section 4.6 for complete detail.

8. Proof of Concept Scope

8.1 PoC Boundary

Dimension	PoC (Phase 1)	Phase 2 post-PoC
Agent count	50–100 agents, one region	Full 4,000 agent deployment
Enforcement type	Bailiff enforcement visits only	CEO parking, environmental, HCEO
Processing mode	Batch only - overnight by 07:00	TBD : Near real-time priority lane (30 min)
AI models	Base NVIDIA models - zero-shot prompting	Fine-tuned on annotated enforcement data
Physical signals	Camera obstruction (no training data needed)	Proximity violations, lunging, objects thrown
Languages	English only	Welsh language routing
Dashboard	Ops dashboard + agent coaching view	Compliance reports, SAR portal, full admin

8.2 PoC Success Criteria

Metric	Target	How measured
SOP step detection precision	> 85%	Human QA on 10% sampled clips
Escalation false positive rate	< 10%	Supervisor review of all flagged clips
ASR word error rate (clean audio)	< 15% WER	Manual transcript comparison on 50 clips
Batch processing completion	All clips scored by 07:00	System log timestamps
Supervisor usefulness rating	> 70% find scores credible	Post-PoC supervisor survey
Score explainability coverage	100% scores have step evidence	Automated schema validation

9. Example Platform Outputs

The following wireframes show expected PoC output format. An interactive HTML version is provided separately.

9.1 Scenario A - Daily Batch Dashboard

KPI Summary Panel

Clips processed	Avg Q1 score	Escalation flags	Vulnerability flags	Require review
47	**76 / 100**	**3 medium · 1 high**	**2 detected**	**5 clips**

Priority Review Queue

Agent	Visit time	Q1 score	Q2 flag	Reason
EA-0042	07:14	**54 / 100**	**HIGH - threat language**	Threat language at 03:42 (agent). Steps 3, 5 missed. Litigation Risk Indicator: HIGH.
EA-0087	14:22	68 / 100	MED - raised voices	Resident-initiated escalation 06:15. Vulnerability flag - elderly visual.
EA-0031	09:05	71 / 100	MED - distress	Distress indicators at 02:30. Vulnerability assessment not conducted.
EA-0019	16:40	62 / 100	MED - 3rd party	Hostile third party detected at 04:10. Agent departed appropriately.
EA-0055	11:30	78 / 100	LOW - camera	Camera angle obstruction at 01:45 for 22 seconds.

9.2 Scenario B - Single Clip Analysis

KPI Summary Panel - EA-0042

Q1 - Procedural	Q2 - Escalation	Q3 - Outcome	Litigation Risk Indicator
54 / 100 FAIL - steps 3, 5 missed	**HIGH** Threat language - agent 03:42	**Entry refused** No goods levied	**HIGH** Supervisor review required

SOP Compliance Checklist - Q1

	SOP step	Status	Timestamp	Evidence
✓	Recording announced	**Detected ✓**	00:08	"This visit is being recorded..."
✓	Identity confirmed - name + cert	**Detected ✓**	00:22	"I am James Cole, certificate 4421..."
✓	Debt amount stated	**Detected ✓**	01:05	"...outstanding balance of £1,840..."
✗	Legal authority explained	**NOT DETECTED ✗**	-	Phrase not found in transcript. −15 pts.
✓	Peaceful entry respected	**Detected ✓**	Visual	Agent posture neutral. Door respected.
✗	Payment options offered	**NOT DETECTED ✗**	-	No payment phrase detected. −15 pts.
⚠	Vulnerability assessed	**UNCERTAIN ⚠**	02:14	Partial - one question. Confidence 0.64.

Escalation Evidence - Q2

Timestamp	Signal type	Speaker	Confidence	Evidence
03:42	**Threat language**	**Agent**	0.91	"You won't like what happens next..."
03:55	Raised voices	Both	0.88	Volume spike - agent voice first. Origin: agent-initiated.
05:10	De-escalation attempt	Agent	0.79	"Let me explain this calmly..."
07:30	Visit ended	-	-	Agent departed. Outcome: entry refused.

10. Scoring Logic

10.1 Scoring Decision Flow

The scoring engine applies rule-based weighted logic for Q1, a severity classifier for Q2, and an outcome classifier for Q3. All three feed into the compliance_review_priority indicator, which maps to the Litigation Risk Indicator displayed on the dashboard.

Figure 6 - Scoring logic decision flow

Figure 6: Q1 SOP compliance (left), Q2 escalation and safety (centre), and Q3 visit outcome (right) scoring trees. All three fuse at the scoring engine into compliance_review_priority. Discriminatory language triggers critical classification regardless of other scores.

11. Training Data Strategy

No publicly available labelled civil enforcement BWC dataset exists. The following three-phase strategy addresses the single most important technical risk in the project.

Phase	Timing	Approach
Phase 1	PoC (weeks 1–10)	Zero-shot inference on base NVIDIA models. Accept ~70–75% precision. Use for discovery and baseline - not operational decisions.
Phase 2	Post-PoC (months 3–6)	Annotate 200–500 hours of real Marston footage using Label Studio / AWS Ground Truth. Labels: SOP steps, escalation events, vulnerability indicators. Fine-tune Riva UK-EN and NeMo.
Phase 3	Ongoing	Supervisor overrides feed back as training labels. Incremental fine-tuning. Model performance tracked per SOP step over time.

12. Risk Register

#	Risk	Likelihood	Impact	Mitigation
R1	No labelled training data from Marston	Medium	High	Three-phase training strategy. PoC zero-shot - accuracy gap acceptable for discovery.
R2	Agent resistance to AI monitoring	Medium	High	Agent comms plan, self-view access, dispute process, coaching-not-surveillance framing, union engagement.
R3	Poor outdoor audio degrading ASR	High	Medium	Zone A2 audio pre-processing. SNR gate routes poor clips to manual review. Expected in 15–25% of clips.
R4	High false positive rate eroding supervisor trust	Medium	High	NeMo Guardrails 0.7 threshold + normalisation layer confidence propagation. Human-in-loop for all consequential decisions. Agent dispute process.
R5	Smart Shuttle / Huddle API access not granted	Medium	High	Pre-PoC critical dependency gate. Fallback: manual footage export workflow until API access confirmed.
R6	ECB standards change during development	Medium	Low	SOP config engine allows updates without code deployment. Change preview validates impact before activating.

13. Open Questions for Marston

#	Question	Priority	Impacts
1	Does Smart Shuttle / Huddle expose an API for programmatic footage access, or will footage need to be exported manually?	**Critical**	Zone A ingestion
2	Will Marston provide access to sample footage (minimum 500 clips) for PoC model evaluation?	**Critical**	AI accuracy baseline
3	Can we obtain the current SOP document for bailiff enforcement visits (all three enforcement stages)?	**Critical**	Zone B SOP config
4	Who is the Data Controller - Marston, the council client, or jointly? This determines the DPIA scope.	**Important**	DPA / DPIA scope
5	How many agents and which region for the PoC cohort?	**Important**	PoC sizing
6	What does success look like to Marston at the end of the PoC? Measurable outcome agreed in advance.	**Important**	Success criteria
7	Is there an existing Azure AD / Entra ID for SSO integration with the dashboard?	Nice to know	Dashboard auth

14. Glossary

Term	Definition
ASR	Automatic Speech Recognition - converting spoken audio to text
BWC / BWV	Body-Worn Camera / Body-Worn Video
CEO	Civil Enforcement Officer - council parking enforcement officer
Compliance review priority	Internal schema field name for the risk indicator computed from Q1+Q2. Dashboard display label: Litigation Risk Indicator.
CPIA 1996	Criminal Procedure and Investigations Act 1996 - governs criminal evidence disclosure
CRAR	Commercial Rent Arrears Recovery - enforcement mechanism for commercial landlords
DPA 2018	Data Protection Act 2018 - UK data protection legislation
DPIA	Data Protection Impact Assessment - mandatory under UK GDPR Article 35 before deploying high-risk processing
DPO	Data Protection Officer
EA / Enforcement Agent	Formerly known as bailiff - certified officer who enforces court-issued warrants
ECB	Enforcement Conduct Board - independent oversight body for the enforcement sector
HCEO	High Court Enforcement Officer - authorised by Lord Chancellor to enforce High Court writs
NeMo Guardrails	NVIDIA AI output safety layer - confidence thresholding and hallucination detection
Normalisation layer	Detection metadata normalisation layer - converts heterogeneous AI engine outputs into a unified timestamp-indexed event graph
PCN	Penalty Charge Notice - civil parking or traffic fine
REIP	Regulated Evidence Intelligence Processing - this platform's category definition
SAR	Subject Access Request - GDPR right to access personal data
SNR	Signal-to-Noise Ratio - measure of audio quality (speech vs background noise)
SOP	Standard Operating Procedure - mandatory steps an enforcement agent must follow
TCoG Regs 2013	Taking Control of Goods Regulations 2013 - primary legislation governing enforcement agent visits
Triton	NVIDIA Triton Inference Server - serves multiple AI models concurrently on GPU
VAD	Voice Activity Detection - identifies speech vs non-speech segments in audio
WER	Word Error Rate - metric for ASR accuracy (lower = better)

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